Shengran Guo

Phone: (667) 434-1423 **Email:** sguo27@ncsu.edu

Address: 401 the Greens Cir, Raleigh, NC, USA

EDUCATION

North Carolina State University
Ph.D. in Computer Science — Aug. 2025 - May 2030 (Expected)
GPA: 3.9/4.0 **Advisor:** Dr. Min Chi

Johns Hopkins University
M.S. in Computer Science — Aug. 2023 - May 2025
GPA: 3.7/4.0 **Advisor:** Dr. Philipp Koehn

Sino-British Collaborative Education Programme — Sep. 2019 - Jun. 2023
BSc (Honours) in Software Engineering, ***Oxford Brookes University (OBU)***
GPA: 83.60/100.00 **(First Class)**
*Top **1**% of the cohort of 95 students*
B.Eng. in Software Engineering, ***Chengdu University of Technology (CDUT)***

RESEARCH INTERESTS

Machine Learning, Data Mining and Generative AI; Reinforcement Learning and Apprenticeship Learning; LLM Inference and Alignment; AI for Education and Healthcare.

INTERNSHIP EXPERIENCE

Amazon — Seattle, WA (Remote)
Machine Learning Engineer Intern — May 2024 - Aug. 2024

- Developed a credit-default prediction model using feature engineering, Random Forest, and LightGBM on **60,000+ samples**, improving ROC AUC from **0.78 to 0.87**.
- Built a search system for **100,000+ GitHub repositories** using API-based data collection, NLP keyword extraction, and clustering for automated topic generation, improving topic prediction accuracy by **22%** and reducing manual tagging workload by **25%**.
- Optimized a machine learning pipeline with PCA-based dimensionality reduction and grid search for LightGBM hyperparameter tuning, reducing processing time by **70%**.

RESEARCH EXPERIENCE

Prof. Min Chi, NCSU

AI-Assisted Text Detection and Cognitively Supervised Automated Grading — Nov. 2025 - Jul. 2026

- Developed **AICG**, a RoBERTa-based multi-task framework combining answer-level grading with Q-matrix-guided skill mastery and learning-gain supervision.
- Built an **AI-Assisted Text Detector** using supervised contrastive learning and a linear classifier, achieving **96.0% F1** on a held-out evaluation set.
- Integrated calibrated AI-assistance probabilities as soft reliability weights for cognitive supervision, reducing grading MAE by an additional **4.2%** over the unweighted cognitive model.
- Evaluated on **49,402 instructor-graded responses** across 19 semesters using expanding-window splits and five random seeds; reduced MAE by **50.3%** and improved quadratic weighted kappa from **0.798 to 0.927** versus RoBERTa regression.

Hierarchical Imitation Learning and Contrastive Reward Modeling — May 2026 - Aug. 2026

- Contributed to **CCHAL**, a hierarchical apprenticeship learning framework for learning decision-making policies from imperfect human demonstrations in healthcare and education.
- Contributed to a **Bradley–Terry pairwise reward-learning module** that contrasts successful and failed trajectories, incorporating decision-level weights into inverse reinforcement learning.
- Conducted benchmark and ablation experiments across **four healthcare and education datasets** with seven baselines and 10 random train/test splits, achieving up to a **17.0% relative improvement** in Jaccard similarity over the strongest baseline.

Assessment Reliability in the Generative AI Era May 2026 - Sep. 2026

- Contributed to **TRUST**, a filtering framework addressing unreliable assessment signals in AI-mediated learning environments by identifying discrepancies between student behavior and assessment outcomes.
- Implemented **Transformer-based** assessment-consistency models with Gaussian heads to predict pre- and post-test scores and uncertainty, using K-fold cross-fitting to generate out-of-sample predictions.
- Evaluated trajectory filtering's impact on offline reinforcement learning; a policy using **19.9% fewer trajectories** showed similar post-test scores and reduced average tutoring time from **79 to 60 minutes**.

Research Collaboration with Dr. Ruikun Luo, University of Macau Mar. 2026 - Sep. 2026
Inference-Time LLM Alignment and Mechanistic Interpretability

- Used **RCM causal scans** to identify preference-supporting and opposing LLM computations.
- Implemented **DHSA activation controllers** with **256–16,384 trainable parameters** for detachable alignment of frozen LLMs through localized interventions during autoregressive token generation.
- Evaluated transfer and ablations across **three alignment tasks**; CAST-enhanced DPO raised GPT-J summarization win rates against human references from **62.11% to 68.55%**.

Prof. Philipp Koehn, JHU Apr. 2024 - Apr. 2025
Cross-lingual Knowledge Transfer in LLMs

- Benchmarked **Llama-3** on translation and question answering across **six languages**.
- Built **two multilingual code-switched datasets** from 5,000 English stories and MMLU questions.
- Fine-tuned Llama-3 on the code-switched datasets to improve cross-lingual knowledge transfer, increasing translation **BLEU from 43 to 49** and **QA accuracy by 5%**.

Prof. Hua Xu, Tsinghua University May 2023 - Aug. 2023
Enhancing Multimodal Capabilities of LLMs

- Integrated ViT and Llama-2 through cross-modal attention, aligning visual and language representations.
- Fine-tuned the model on **COCO** and **VQA** and used beam search for caption generation, improving VQA accuracy by **3%** and captioning BLEU by **5%**.
- Implemented **CLIP-like contrastive pre-training** to align image/text embeddings for retrieval.

SELECTED PUBLICATIONS

- **Shengran Guo**, Min Chi, Feifei Qiao, Varad Patwardhan, "Beyond Response Scoring: An AI-aware Cognitive Supervision Framework for Automated Mathematical Short Answer Grading", **under second-round review at AAAI 2027**.
- Md Mirajul Islam, **Shengran Guo**, Xi Yang, Min Chi, "Critical Contrast Hierarchical Apprenticeship Learning from Imperfect Demonstrations", **submitted to KDD 2027**.
- Fansheng Zhang, **Shengran Guo**, Zexiao Wang, Liang Yuan, Jiyuan Chen, Ruikun Luo, "Direct Hidden-State Alignment: Mapping and Controlling Preference Expression in LLMs", arXiv:2609.33298, 2026; **submitted to ICLR 2027**.
- Rajesh Debnath, **Shengran Guo**, Md Mirajul Islam, Min Chi, "TRUST: Trace-Based Recovery of Unbiased Student Trajectories for Robust Pedagogical Policy Induction", **submitted to LAK 2027**.
- **Shengran Guo**, "Application and Impact of Position Embedding in Natural Language Processing", **ICEIB 2024**, Pages 246–250, 2024.

PROFESSIONAL SKILLS

Programming: Python, Java, C/C++, SQL, MATLAB; familiar with R.
ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn.
LLMs & NLP: Transformer Architectures, LLM Fine-tuning, AI-Generated Text Detection, Activation Steering, Causal Interventions, Cross-lingual Transfer and Multimodal Alignment.
Machine Learning: Offline and Inverse RL, Preference Optimization (DPO), Pairwise Reward Modeling, Supervised Contrastive Learning, Multi-task Learning, and Uncertainty Estimation.
Research Computing & Tools: Linux, Git, Slurm, GPU computing, AWS, MySQL, and Jupyter